Course Notes

TOPICS IN ALGEBRA: GROUP THEORY

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Chapter 1

Lecture 1

Lie algebras were originally introduced by Sophus Lie in the 1870s to study the concept of continuous transformation groups. Lie's motivation was to develop a theory for differential equations analogous to Galois theory for algebraic equations. He discovered that the infinitesimal transformations of a continuous group form a linear space equipped with a bracket operation, now known as a Lie algebra. This linearization allows complex global geometric problems to be translated into algebraic problems, making them more tractable.

Let F be a field (mostly $\operatorname{char} F = 0$, even $F = \mathbb{C}$).

Definition 1.0.1. *A linear algebra A is a vector space over F, with a binary bilinear operation:*

- *binary operation: $A \times A \to A$, $(a, b) \mapsto ab$.*
- *bilinear: this operation is linear in each variable. In orther word, $\forall \alpha \in F$ and $a, a', a'', b, b', b'' \in A$, we have:*

$$(\alpha a)b = a(\alpha b) = \alpha(ab)$$
$$(a' + a'')b = a'b + a''b$$
$$a(b' + b'') = ab' + ab''$$

The algebra A is associative if $\forall a, b, c \in A$, $(ab)c = a(bc)$.

Example 1.0.2. *$M_n(F)$, the algebra of $n \times n$ matrices over F.*

Example 1.0.3. *Let V be a vector space, $\operatorname{End}_F(V) = \{$linear transformations $\varphi : V \to V\}$ is an associative algebra with the composition operation$(\phi\psi)(v) = \phi(\psi(v))$. If $\dim V < \infty$, when we fix a basis, then we fet a matrix representation of φ in our fix basis.*

The most imprtant thing everyone shoul know is Wedderburn's theorem.
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Definition 1.0.4. *Suppose F is algebraically closed. An algebra A is simple if*

1. $\nexists\ I \lhd A$ *and* $I \neq 0$.
2. $A \cdot A \neq (0)$.

Corollary 1.0.5. *Any simple finite dimensional associative algebra is isomorphic to $M_n(F)$ for some n.*

Corollary 1.0.6. *Radical*

1.1 Where do Lie algebras come from?

Definition 1.1.1. *A Lie algebra is a vector space L over a field F, with a binary operation $[a, b] \in L$, such that*

1. $[a, b]$ *is bilinear.*
2. $[a, b] = -[b, a]$.
3. $[a, [b, c]] + [c, [a, b]] + [b, [c, a]] = 0$, *this is called Jacobi identity.*

1.1.1 They come from associative algebras

Let A be an associative algebra. $\forall a, b \in A$, define $[a, b] = ab - ba$. Check that $(A, [\ ,\ ])$ is a Lie algebra:

Proof. Bilinearity and skew-symmetry are clear. For Jacobi identity:

$$\begin{aligned}
&[a, [b, c]] + [b, [c, a]] + [c, [a, b]] \\
&\qquad = a(bc - cb) - (bc - cb)a + b(ca - ac) - (ca - ac)b + c(ab - ba) - (ab - ba)c \\
&\qquad = abc - acb - bca + cba + bca - bac - cab + acb + cab - cba - abc + bac \\
&\qquad = 0.
\end{aligned}$$

□

We denote this new algebra $(A, [\ ,\ ])$ by $A^{(-)}$.
Suppose that $L \subseteq A$ and $[L, L] \subseteq L$. Then $L \leq A^{(-)}$。

Example 1.1.2. *$M_n(F)^{(-)} = \mathfrak{gl}(n)$. We call it the general linear Lie algebra.*
$[L, L] = \{\sum_i [a_i, b_i]\} \trianglelefteq L$. Clearly $[\mathfrak{gl}(n), \mathfrak{gl}(n)] \neq \mathfrak{gl}(n)$ and for $n \geq 2$, $[\mathfrak{gl}(n), \mathfrak{gl}(n)] \neq 0$.

Example 1.1.3. *Focus on* $[\mathfrak{gl}(n), \mathfrak{gl}(n)]$*, all matrices are of trace 0. This is called the special linear Lie algebra, denoted by* $\mathfrak{sl}(n)$*.*

Definition 1.1.4. Z *is the center of Lie algrbra* L *if* $Z = \{z \in L | [z, L] = 0\}$*.*

We can also find Lie algebras from the involution of an associative algebra. Let A be an associative algebra. $* : A \to A$ is a linear transformation. $*$ is called a an involution if $(a^*)^* = a$ and $(ab)^* = b^* a^*$, $\forall a, b \in A$.

Example 1.1.5. *The Quatanion algebra is an associative algebra with the involution* $a \mapsto \overline{a}$*. It is 4 dimensional over the base field with standard basis* $1, i, j, k$*. We use* $\mathbb{H}$ *to denote the Quatanion algebra when we choose* $\mathbb{R}$ *as the base field. i.e.* $\mathbb{H} = \{a_0 1 + a_1 i + a_2 j + a_3 k | a_0, a_1, a_2, a_3 \in \mathbb{R}\}$*.*

Chapter 2

Lecture 2

Recall last time:

1. Matrix Lie algebras, e.g. $\mathfrak{sl}(n)$. Consider involution $* : M_n(F) \to M_n(F)$ skew-symmetric matrices.

2. Derivation, $\mathrm{Der}(A)$. e.g. $A = F[t_1, \ldots, t_n]$, $\mathrm{d} = \sum_{min}^{max} f_i()\partial$

A non-associative algebra, $\mathrm{d} : x \mapsto [a, x]$ is a derivation.

Let L a Lie algebra, inner derivation

Q: d a derivation and exp(d). It need convergence.

Example 2.0.1. $\mathrm{d} = t_1 t_2 \frac{\partial}{\partial t_1}$.

3. Brackets:

2.1 Engel Theorem

L^n: span of products of elements of length n. Recall $\mathrm{ad}([a, b]) = [\mathrm{ad}(a), \mathrm{ad}(b)]$. $L \to \mathrm{End}_F(L)^{(-)}, a \mapsto \mathrm{ad}(a)$.

$[I, J]$ is again an ideal.

Decending chain:

$$L = L^1 \supseteq L^2 \supseteq \ldots$$

L is nilpotent if....

eg: strict upper matrix

Another decending chain: $L^{[1]} = [L, L]$, $L^{[n+1]} = [L^{[n]}, L^{[n]}]$

L is solvable if .....

e.g. 2 dimentional sovable non nilpotent Lie alg.

A associa. $L \subseteq A^{(-)}$. Suppose $A =< L >$ as an associa alge. Then A is an enveloping algebra of L.

Theorem 2.1.1. *Let A be a finite dimensional associative algebra, $L \subseteq A^{(-)}$, $A =< L >$. $\exists n \geq 1$, $\forall a \in L$, $a^n = 0$. Then A is nilpotent.*

We say that a subset $S \subseteq L$ is a Lie subset if $\forall a, b \in S$, $[a, b] \in S$.

Theorem 2.1.2. *Let A be a finite dimensional associative algebra, $L \subseteq A^{(-)}$, $A =< L >$. Let S be a Lie subset of L. $L = \mathrm{span}(S)$, $\forall a \in S$, $a^n = 0$.*

Choose a maximal Lie subset $S_1 \subset S$ such that the associative $< S_1 >$ is nilpotent. So $\exists n$ such that $S_1 \ldots S_1 = (0)$ (n times) (existance?)

claim $\mathrm{ad}(S_1) \ldots \mathrm{ad}(S_n) = (0)$.